Converting Units

Bigger unit to smaller: multiply. Smaller to bigger: divide.

Metric units go by 10, 100 and 1,000

1 km = 1,000 m	1 kg = 1,000 g
1 m = 100 cm	1 L = 1,000 mL
1 cm = 10 mm	1 m = 1,000 mm

2.5 m in cm: a bigger unit, so more of the smaller one

$2.5 \times 100 = 250$ Times 100: the digits move 2 places left.

2.5 m = 250 cm

350 cm in m: a smaller unit, so fewer of the bigger one

$350 \div 100 = 3.5$ Divide by 100: the digits move 2 places right.

Feet and inches: what is left over

50 in = 4 ft 2 in 12 in = 1 ft. 50 ÷ 12 = 4, and 2 left over.

Your turn!

4.6 kg = 4,600 g

For teachers

Goal: convert both ways inside one system, with decimals in metric (5.MD.1). It uses packet 73's digit moves. Say “Bigger unit, so I need more of the small ones: multiply.”

Converting Units: Practice

Bigger to smaller: multiply. Smaller to bigger: divide.

Part 1 Bigger to smaller Multiply by 10, 100 or 1,000.

1)	2.5 m = 250 cm	2)	3.4 kg = 3,400 g	3)	1.75 L = 1,750 mL
4)	6.2 km = 6,200 m	5)	4.8 cm = 48 mm	6)	0.6 m = 60 cm
7)	2.05 kg = 2,050 g	8)	0.9 L = 900 mL	9)	7.3 m = 7,300 mm
10)	12.5 km = 12,500 m	11)	0.35 m = 35 cm	12)	5.4 cm = 54 mm

Converting Units: Practice (continued)

Part 2 Smaller to bigger Divide by 10, 100 or 1,000.

13) 350 cm = 3.5 m

14) 2,600 g = 2.6 kg

15) 750 mL = 0.75 L

16) 4,200 m = 4.2 km

17) 85 mm = 8.5 cm

18) 1,250 g = 1.25 kg

19) 60 cm = 0.6 m

20) 3,400 mm = 3.4 m

21) 500 m = 0.5 km

22) 1,900 mL = 1.9 L

23) 47 mm = 4.7 cm

24) 820 cm = 8.2 m

Converting Units: Practice (continued)

Part 3 Left over No help this time.

25) 50 in = 4 ft 2 in

26) 75 min = 1 hr 15 min

27) 40 oz = 2 lb 8 oz

28) 11 ft = 3 yd 2 ft

29) 100 in = 8 ft 4 in

30) 150 sec = 2 min 30 sec

31) 30 hr = 1 day 6 hr

32) 45 days = 6 weeks 3 days

Part 4 Which is more? No help this time. Change to the same unit first.

33) 1.2 kg > 1,150 g

34) 3.5 m < 400 cm

35) 0.8 L < 850 mL

Converting Units: Practice (continued)

Part 4 Which is more? (continued)

36) 2.3 km [=] 2,300 m

37) 45 mm [>] 4.4 cm

38) 1.06 m [<] 160 cm

39) 700 g [=] 0.7 kg

40) 2.5 L [>] 2,050 mL

41) 9 cm [>] 85 mm

Part 5 Add mixed units No help this time. Change to the small unit first.

42) 1.5 m + 30 cm = [180] cm

43) 2 kg + 450 g = [2,450] g

44) 1.2 L + 350 mL = [1,550] mL

45) 3 km + 800 m = [3,800] m

46) 2.4 cm + 5 mm = [29] mm

47) 0.5 kg + 175 g = [675] g

Converting Units: Practice (continued)

Part 6 Bigger to smaller

48) 9.3 m = 9,300 mm

49) 11.1 m = 1,110 cm

50) 1.2 L = 1,200 mL

51) 8.5 kg = 8,500 g

52) 6.9 kg = 6,900 g

53) 1.2 m = 1,200 mm

54) 6.8 L = 6,800 mL

55) 9.9 km = 9,900 m

56) 2.6 km = 2,600 m

57) 1.4 km = 1,400 m

58) 4.7 m = 470 cm

59) 8.4 cm = 84 mm

Converting Units: Practice (continued)

Part 6 Bigger to smaller (continued)

60) 1.7 km = 1,700 m

61) 8.4 m = 840 cm

62) 9.4 cm = 94 mm

63) 6.2 m = 6,200 mm

64) 10.9 m = 10,900 mm

65) 6.1 L = 6,100 mL

66) 3.4 km = 3,400 m

67) 6.6 cm = 66 mm

68) 1.4 cm = 14 mm

69) 12.2 L = 12,200 mL

70) 2.9 kg = 2,900 g

71) 7.8 km = 7,800 m

Converting Units: Practice (continued)

Part 7 Smaller to bigger

72) 750 g = 0.75 kg

73) 890 g = 0.89 kg

74) 250 mL = 0.25 L

75) 500 cm = 5 m

76) 700 m = 0.7 km

77) 210 mm = 21 cm

78) 700 g = 0.7 kg

79) 960 mL = 0.96 L

80) 690 mm = 69 cm

81) 320 mm = 32 cm

82) 520 mL = 0.52 L

83) 230 g = 0.23 kg

Converting Units: Practice (continued)

Part 7 Smaller to bigger (continued)

84) 610 m = 0.61 km

85) 430 mm = 43 cm

86) 730 m = 0.73 km

87) 920 mL = 0.92 L

88) 370 m = 0.37 km

89) 460 mm = 0.46 m

90) 450 g = 0.45 kg

91) 460 cm = 4.6 m

92) 970 mL = 0.97 L

93) 120 g = 0.12 kg

94) 220 g = 0.22 kg

95) 610 g = 0.61 kg

Converting Units: Practice (continued)

Part 8 Remember? Problems from earlier packets.

96) $(16 - 7) \times 4 =$ **36**

97) Start at 3, add 9: 3 12 **21**

98) $\frac{1}{6} \div 3 = \frac{1}{18}$

99) $8 \div \frac{1}{3} =$ **24**

100) $5.6 \times 1{,}000 =$ **5,600**

101) $\frac{3}{8} \times \frac{2}{3} = \frac{6}{24}$

Part 9 Word problems Change to the same unit first.

102) A ribbon is 2.4 m long. Mia cuts off 60 cm. How many cm are left?

Number sentence **240 − 60 = 180**

Answer **180** cm

Draw or show your work

103) A bag of flour weighs 2.5 kg. A cake uses 300 g. How many grams are left after 3 cakes?

Number sentence **2,500 − 3 × 300 = 1,600**

Answer **1,600** g

Draw or show your work

104) Leo drinks 250 mL of milk each day. How many liters does he drink in 6 days?

Number sentence **6 × 250 = 1,500 mL**

Answer **1.5** L

Draw or show your work

Converting Units: Practice (continued)

Part 9 Word problems (continued)

105) A race is 5 km. Zoe has run 3,750 m. How many more meters does she need to run?

Number sentence **5,000 − 3,750 = 1,250**

Answer **1,250** m

Draw or show your work

106) Challenge: A shelf is 1.2 m long. How many 15 cm books fit on it side by side?

Number sentence **120 ÷ 15 = 8**

Answer **8** books

Draw or show your work

Check-In: Converting Units

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 Bigger to smaller

1) 2.25 L = 2,250 mL

2) 7.2 cm = 72 mm

3) 4.5 km = 4,500 m

Part 2 Smaller to bigger

4) 450 cm = 4.5 m

5) 3,200 g = 3.2 kg

6) 2,800 m = 2.8 km

Part 3 Left over

7) 38 in = 3 ft 2 in

8) 90 min = 1 hr 30 min